



B.M.S. College of Engineering
Department of Information Science and Engineering
Capstone Project Phase - 1 (23IS6PWPW1)
AY:2025-26
SYNOPSIS

Batch No: 38	Project Title: Dinacharya Automation Using LLM-based Autonomous Agents: A Dosha-aware Multimodal Personalization Framework
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Name of the Guide:	Harini S
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Group members:

Sl. No.	USN	NAME
1.	1BM23IS243	Siddharth Amdal
2.	1BM23IS261	Swapnil Patidar
3.	1BM24IS418	R V Kirrthanaa
4.	1BM23IS237	Shreyansh Kumar Singh

Abstract:

Dinacharya, a classical Ayurvedic concept translating to daily regimen, prescribes time- and constitution-sensitive practices intended to maintain physiological balance and promote long-term health. This project proposes the design, development, and evaluation of an LLM-based autonomous agent that generates, schedules, and adapts personalized dinacharya recommendations through multimodal sensing and an explicit Ayurvedic knowledge base. The agent synthesizes user profile data (prakriti and vikriti assessments), wearable signals (sleep, heart rate variability, activity), smartphone context (calendar, location), and self-reported symptoms into a closed-loop personalization pipeline. Core components include:

- (1) A Dosha mapping and feature extraction module that translates traditional Ayurvedic assessments into computable phenotypes;
- (2) An LLM planner and NLU module for natural language generation, explanation, and dialogue;
- (3) A retrieval-augmented knowledge base to ground recommendations and prevent hallucination;
- (4) A rule-based safety checker to enforce clinical constraints; and
- (5) A behavior change engine that schedules nudges and reminders with adaptation based on adherence and physiological proxies.

Implementation will use modern ML and engineering tools (PyTorch/Hugging Face models, LangChain for agent orchestration, FAISS/Weaviate for vector retrieval, and a mobile client for deployment). Evaluation includes a pilot deployment (n=20) to measure adherence, subjective usability, and objective physiological proxies (sleep duration, HRV). Statistical analysis and ablation studies will quantify the contribution of multimodal grounding and dosha personalization. The project aims to demonstrate the feasibility and preliminary efficacy of a dosha-aware, agentic approach to daily routine automation, while documenting safety, privacy, and validation pathways for future clinical translation.

1. INTRODUCTION:

1.1 Background

Dinacharya denotes a daily regimen prescribed within Ayurvedic medicine. It includes specific temporal and procedural recommendations for waking, hygiene, elimination, meals, exercise, rest, and sleep. These recommendations are framed by individual constitution (prakriti), current imbalance (vikriti), seasonality, and circadian considerations. Contemporary digital health systems and consumer wearables have enabled continuous monitoring of physiological signals and behavior, creating an opportunity to operationalize traditional guidance in a personalized, data-driven manner. Simultaneously, advances in large language models (LLMs) and agentic architectures permit flexible, conversational, and tool-enabled automation for complex, longitudinal tasks.

1.2 Motivation

Adherence to beneficial daily routines remains low for many individuals due to lack of personalization, poor integration with daily context (work schedules, travel), and limited feedback loops. Ayurvedic prescriptions are rich in personalized nuance but are not directly actionable in their textual form. An agentic system can translate Ayurvedic knowledge into daily schedules adapted to the user's physiology, preferences, and constraints, and can maintain a closed-loop adaptation mechanism to improve adherence and outcomes. This synthesis of traditional knowledge and modern sensing offers potential scientific and public health value.

1.3 Objectives

- To design an LLM-based autonomous agent that constructs personalized, temporally organized dinacharya plans grounded in a vetted Ayurvedic knowledge base.
- To implement multimodal data fusion combining wearables, self-report, and contextual signals for closed-loop adaptation.
- To ensure safety by integrating a rule-based checker and expert-vetted constraints to avoid harmful or contraindicated recommendations.
- To evaluate feasibility and preliminary effectiveness through a pilot study measuring adherence, usability, and physiological proxies.

1.4 Application

Primary applications include: consumer wellness assistants that operationalize dinacharya; clinician-assisted tools for lifestyle interventions; and research platforms for studying the interactions between behavioral schedules and physiological outcomes. The system is designed primarily for non-acute, preventive settings and explicitly excludes diagnosis or acute medical triage.

2. PROBLEM DEFINITION

Modern wellness recommendations are generic and fail to incorporate traditional personalized frameworks such as dinacharya. Existing digital solutions rarely combine culturally grounded domain knowledge, multimodal sensing, and agentic planning. The problem addressed is: how to construct a safe, explainable, dosha-aware agent that translates Ayurvedic daily regimen into adaptive, contextually feasible plans, and to evaluate whether such an agent improves adherence and physiological proxies compared to baseline conditions.

3. GANTT chart

TASK	FEBRUARY		MARCH		APRIL		MAY	
	Weeks 1 - 2	Week 3 - 4	Week 1 - 2	Week 3 - 4	Week 1 - 2	Week 3 - 4	Week 1 - 2	Week 3 - 4
Literature Review & Problem Definition								
System Design								
Data Collection & Integration								
Model Evaluation								
Agent Development								

Yes	No
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Synopsis: Accepted

Signature of the Team members:

Signature of the Guide: